A Research Agenda for the Two Moves: Companion Papers to the Book

Research agenda v0.37, Project secret3, August 11, 2026

This agenda lists concrete research lines connecting the two-combinator calculus to numbers, computation, logic, and physics, in the pattern the primes work established: an exact question, an engine or a proof campaign, and results promoted to compiled theorems wherever they can be. Each line names a proposed paper, candidate theorems or measurements with signatures where possible, method, effort, and the book chapter it serves. Two seed theorems ship with this agenda and are already compiled in the proof module (part K): sumValue, the additive companion of Theorem B, and boxLen, the exact cost of the canonical multiplication. Effort grades: S is a session, M is days, L is a campaign.

1. Numbers

N1, strong normalization of R_arith. Question: does every R_arith reduction terminate, the standing open obligation of paper one? Method: the skeleton homomorphism onto a free Peano signature and a recursive path ordering, paper-level first, formal later. Candidate theorem: SN of the skeleton system; transfer lemma along the homomorphism. Paper: extension of the tree-numbers paper to v0.5. Effort M paper-level, L formal. Serves The Unreachables.

N2, the FTA module. Question: discharge the arithmetic hypothesis of Proposition 4, making F_arith confluence equivalent to unique factorization unconditionally. Method: divisibility, primes, and unique factorization in base-library Idris2. Candidate theorems: euclidLemma, factorizationUnique, and the unconditional Proposition 4. Effort L. Serves The Unreachables.

N3, additive reachability and a Goldbach census. Question: which numerals are reachable as sums of two prime numerals inside the calculus? Status: the reachability characterization is now a theorem, shipped with this agenda: sumStepsNum gives the forward derivation and sumValue : StepsR (sumBox (num m) (num k)) (num n) -> m + k = n closes the backward direction by the value invariant, so a sum box reaches exactly the sum. Next: an engine census of prime-pair reachability over a window, lifted by allBelowSound to a bounded universal theorem relative to a checked run. Paper: 'Addition as Reachability, and a Verified Goldbach Window'. Effort S for the census, done for the theorem. Serves The Unreachables.

N4, the cost of a times table. Question: what does arithmetic cost, exactly, in derivation steps? Status: seeded and compiled: boxLen : rLen (boxReduces m k) = S (m * (S (S k))), the canonical multiplication of m by k takes exactly m times (k + 2) plus one steps. The formula is not symmetric: two times one hundred costs 205 steps, one hundred times two costs 401; the schoolteacher's rule, put the smaller number first, is now a theorem about this derivation. Delivered in full with P1: the weight counters dupMass and eraseMass are compiled with their congruence family, and the closed forms are exact: boxDup, m copies of the numeral of k; boxErase, that numeral once; sumDup and sumErase, zero, addition is linear; tSizeNum, a numeral costs 5n plus 4 cells. Reported in 'The Entropy Budget of Multiplication' (working paper v0.1). Serves The Price of Forgetting and The Price List.

N5, the unreachability spectrum. Question: beyond numerals, which terms are box-unreachable, and what is the classification of R-normal forms? Method: extend the AF family and the inversion machinery; census by engine, classification by proof. Effort M to L. Serves The Unreachables.

2. Computation

C1, the null-model census and autocatalytic sets. Question: does any species in the driven conserving vessel stand above its random-assembly expectation, and does the reaction network contain reflexively autocatalytic, food-generated sets in the Hordijk and Steel sense? Method: equilibrium ensemble estimation from the cleave and condensation flux; the RAF detection algorithm run on the logged reaction network. Delivered in soup paper v0.3: the operational null isolates the chemistry's fingerprint, atom pools enriched three to four fold and the two-atom sector tripled, with three-atom per-species claims honestly retired as below resolution; the RAF question degenerates, as the chemistry has no catalysis, to total food-closure, computed, and the recycling core, measured at roughly half the species. The specified continuation is time-averaged, multi-seed censusing. Serves The Vessel.

C2, the phase diagram of the chemistry. Question: treat the cleavage rate as a temperature and the condensation cap as a pressure; where is the freezing transition, and is there a regime between glass and thin steady state with rich standing crops? Delivered as 'Freezing and Flux in a Conserving Combinator Chemistry' (working paper v0.1): sixteen runs, two seeds, flux non-monotone with its maximum at the balanced melting fraction where diversity also peaks, the mechanism decomposed into falling attempt volume against rising success rate carried by copy availability, the two starvations named, and pressure shifting the boundary with the sweep's records at the tight cap. The refinement sweep and the rule-set generalization are specified. Serves The Vessel and the physics of driven systems.

C3, a verified Kolmogorov atlas. Question: what are the provably minimal SK terms for small behaviors (identity, the projections, composition, the successor on Church numerals), with minimality as a theorem rather than a belief? Method: exhaustive enumeration to size n by the E4 engine, behavioral testing, and the allBelow bridge lifting the sweep to a compiled bounded statement: no term of size at most n has the behavior, and here is the size n + 1 witness. Caution: type-level exhaustion is compile-time heavy; the design reflects runtime sweeps into checked certificates for small n. Paper: 'A Verified Atlas of Shortest Programs'. Effort M. Serves Two Moves and the program-size thread.

C4, sharing against trees. Question: how much duplication waste does tree reduction pay over graph sharing on this arithmetic? Method: instrument a sharing-graph evaluator beside the tree engine; measure on the numeral benchmarks. Effort M. Serves The Answer.

C5, a verified reader. Question: a self-interpreter for the calculus with compiled behavior: a quotation scheme for terms and a universal term U with the theorem that U applied to the quote of t, applied to quoted input, walks to the quote of t's result. Status: stage one is delivered, the quoted alphabet ('A Quoted Alphabet', working paper v0.1): Code and embed, Scott quotation with the workshop pair as its spine, the three-case interface qSCase, qKCase, qAppCase as compiled move chains, and the discipline theorems quoteClosed and quoteInj. The vanishing engine (abs, absCorrect, absGone) stands beside it from chapter two. Stage two is delivered, the verified engine ('The Reader's Engine', working paper v0.1): simStep computes leftmost reduction at the code level, with simSound (every reported step is a real step), simNormal (silence means normal form, a constructed negation over all positions and rules), and simRunSteps (fueled runs are real walks). The engine is the specification. Stage three is underway: movement one is delivered ('The First Eye', working paper v0.1), the bench (sub1NoOcc, absSubSteps, abs2Beta, with quoteClosed discharging every side condition), the internal tags with their cases, the constructor qMk with qMkBeta, and the verified K detector readK, seven theorems covering the code space exactly, every proof a composition of named lemmas. Movement two is delivered ('The Second Eye and the Assembler', working paper v0.1): the bench through arity four (absSubSteps2, absSubSteps3, abs3Beta, abs4Beta, jTag-

Cong), and readS with nine theorems matching the engine's root-S clause exactly, the firing branch assembling the contractum by three internal constructor calls, the share performed on descriptions, congruence-lifted chains rehearsed for the knot. Movement three is delivered ('The Dispatcher', working paper v0.1): rootRead composes the eyes through the tag cases, nine theorems, exact agreement with the engine's root fragment, longest proof four links. Everything below the knot is standing, twenty five behavior theorems across three movements. Movement four is delivered and the campaign is complete ('The Knot: A Verified Self-Reader for the Two-Move Calculus', working paper v0.1): simT tied by self-application with a one-lemma unfolding, and simCorrectJ with simCorrectN establishing agreement with the engine on every code, nine clauses each way, fifth consecutive first-attempt compilation. Sixty two named theorems across the campaign's six papers. C5 is closed; the reader, being a closed term, falls under its own theorems, and The Machine That Runs Itself has its receipt. Paper: 'A Verified Reader'. Effort L. Serves Everything from Two and The Machine That Runs Itself.

3. Logic

L1, the verified coin grown. Question: subject reduction for the Hilbert fragment: if Deriv a t and t steps to u, then Deriv a u. Method: inversion lemmas on MP spines against the two redex shapes; the machinery mirrors the stepRAF campaign. Preservation delivered: derivPreserved and derivSteps are compiled via spine inversion and arrow injectivity, reported in 'A Verified Coin: Type Preservation for the Two-Move Hilbert System' (working paper v0.1), together with the quantity lesson: step relations one rebuilds typed terms from must carry relevant term indices, a refactor the whole module absorbed unchanged. The normalization half is delivered: strongNorm, strong normalization for the coin by the reducibility method with semantic neutrality and cargo descent ('Every Typed Song Ends', working paper v0.1). L1 is closed, preservation and termination both compiled; the garden's sentence is owned, and with stage two's engine the corollary stands in prose: the reader always finishes reading a proof.

L2, no bird sings Peirce. Question: make the underivability of Peirce's law a compiled theorem. Method: a three-valued matrix inside the module: values Fin 3, implication x -> y defined as top if x is at most y and y otherwise; K and S are valid at all nine and twenty seven instantiations, checked by a single Refl over a computed sweep; soundness of Deriv by one induction; the valuation a := 1, b := 0 refutes Peirce. Candidate theorem: peirceNoProof : {t : Term} -> Deriv (((a -> b) -> a) -> a at the refuting instance) t -> Void. Delivered: peirceNoProof and dneNoProof are compiled, universally quantified over proof terms, with the same valuation shutting both doors, and the paper 'No Bird Sings Peirce' (working paper v0.1) reports the construction; the book's Garden chapter now has its sharpest receipt. Serves The Garden.

L3, substructural coins. Question: Deriv variants without K (relevance), without contraction and weakening (linear), with the ledger correspondence to the occurrence laws made formal: derivations in the K-free coin type exactly the terms whose reductions never erase. Effort M. Serves The Aviary and The Price of Forgetting.

L4, the rewinder made operational. Question: extend the term language with a control operator, type it by Peirce's law, and verify Griffin's correspondence in the small: classical proofs as programs that back-track. Effort L. Serves The Garden.

4. Physics

P1, the entropy budget (E7). Delivered with N4 in 'The Entropy Budget of Multiplication' (working paper v0.1). Headline: the disagreement of prices, step count is minimized by putting the smaller factor first, erasure by putting the larger factor first, so under Landauer's principle the fast order is the hot order; at 2 and 100 the fast order erases thirty-six times more material. The Landauer read-

ing is scoped honestly: floors, not bills; payload accounting; unary on purpose, with the positional caveat stated. Remaining in this line: the probe's ledger and instrumented soup-run budgets, effort S, if the vessel chapters want them.

P2, a verified reversible machine. Question: build the BCI fragment as a machine whose steps are provably invertible, Bennett's thesis in combinator clothing. Candidate theorems: backward determinism of the linear step relation on a history-carrying presentation; round-trip lemmas. Paper: 'A Verified Reversible Combinator Machine'. Effort M to L. Serves The Price of Forgetting and The Witness.

P3, the boundary precision piece. Question: state, with full citations and the project's fragment classification from the linearity paper, the correspondence between the quantum refusals (no cloning, no deleting) and the availability of the two moves, including quantum Darwinism's redundant copying as the S-side of classicality. Method: synthesis, no new theorems claimed, the register the program note reserved for it. Paper: 'Where the Two Moves Are Refused'. Effort S to M. Serves The Witness.

P4, thermodynamics of the vessel. Question: joint with C2: read the phase diagram thermodynamically, entropy production per tick from the reaction mix, the frozen glass as the zero-flux state, and the driven steady state's budget. Effort M. Serves The Vessel and The Price of Forgetting.

Line Q: quantum shadows

Q1 (delivered). The two refusals as occurrence conservation ('Two Refusals, One Ledger', working paper v0.1): occStepEq and occRunEq meter every computation; noClone and noDelete state the refusals fragment-free; cloneBill and deleteBill price the exceptions and connect them to the entropy budget's Landauer reading. Serves The Witness (The Two Refusals section) with compiled in-house cousins of the quantum prohibitions.

Q-next (candidates, unranked). The reversible fragment characterized quantitatively (both meters zero on every run against linear definability; prior art trail James and Sabry, information effects). The boundary question for conserving chemistries is answered by reframing ('The Assembly Ledger', working paper v0.1): there is no boundary of permission, there is one law of payment over both vessels, condBonds, kBonds, sBonds, and stepsNeverMint, computation cannot mint structure, every daughter is paid bond for bond, the absorber metered as a furnace (tauMealMint, tauMealBurn); the remainder, which samplers discover metabolic programs, is empirical and is not dressed as a theorem. Newly opened per the author (line C6, the chemist's question): take the vessel's road further toward real atoms and molecules; determine which laws of chemistry are combinatorial theorems under conservation, first target delivered ('Valence: The First Atomic Property', working paper v0.1, module part Z): valence as bond turnover, K a pure sink at two, S turnover neutral at three, the values compiled as the undersaturated species (valuesUndersaturated), and the flow law stepsBondFlow, broken equals formed plus burned along every computation. Second delivery ('Coordination and Turnover: Valence II', working paper v0.1, module part AA): the polarity inversion against shell chemistry named as a correction, the atoms reread as catalytic centers, and the two-regime law compiled as progress (rootStatus: every closed molecule gathers or fires, firing exactly at valence, the fired atom the head atom, condensation climbing one rung per event). The saturation-weighted vessel sweep is thereby upgraded from borrowed heuristic to law-shaped rule and remains the registered tier-two experiment; then reaction-network thermodynamics from the bond ledger, then phase structure as a theorem; the thermodynamic half of valence, stable preference rather than turnover, remains open, its home is energy and the calculus has none yet. Newly opened per the author (line C8, nucleic combinators): N1 delivered ('Nucleic Combinators I: The Complement Involution', working paper v0.1, module part AB, slice-verified): compT with its involution, the four-base alphabet as values in the symmetric saturation zone, basePairing as the K

against S duality letterwise, baseVal for inertness, and replicationExact, fidelity by theorem, the copier complements twice; the asymmetry finding, that complementation ignites the S-holding-two shape into a saturated Kestrel, bounds the alphabet and is reported. Registered: N2, the polymerase campaign, C5-shaped, spec in the paper, strand encoding in the songbook its first task; N3, hybridization and melting on the existing temperature dial, prediction that replication throughput peaks in the busy middle; N4, the codon constructor toward the nest-builder. Newly opened (line M1, infrastructure, required): the shared module hit the sandbox checker's four-gigabyte ceiling at part AB; modularize Proofs.idr into a package, one part per module under public export, top-module check as the one command; until M1 lands, new parts are verified against exact dependency slices, shipped beside the module. Line C8 addendum ('The Other Alphabet', research note v0.1, module part AC, slice-verified): BCWI as composite bases assessed, a legitimate same-zone alphabet, distinct and inert, that the canonical duality uniformly ignites, so no found pairing exists; the pairable zone is characterized exactly, heads at saturation zero or one, and the N1 alphabet is its minimal symmetric choice, a design fact binding future alphabet enlargements. Line C8, the chiral branch ('The Chiral Base', working paper v0.1; Chiral.idr, standalone seed, first-try, whole-file check as the one command): the designed answer to the author's base question, derived from the program's own requirements; four atoms in two hands, complementation the mirror, and the crown theorem mirrorStep, pairing as an automorphism of the laws; the unbidden gifts, antiparallelism as a corollary of reflection, the A and T pair at valence two against G and C at three matching the two hydrogen-bond counts, and the transitions-against-transversions metric native to the alphabet; per-hand completeness cited; CH1 delivered ('Chiral II', working paper v0.1; Chiral.idr at 187 lines, whole-file check): strandNoStep by the ChainShape view, antiStrandNoStep by automorphism transport alone, and mirrorChain, antiparallelism as an equation, the mirror of a sense strand is the complement on the opposite spine. CH2 closed as delivered, standalone, Chiral.idr at three hundred and eighty-four lines checking whole: the polymerase's constitution compiled in four movements, legibility with the checker-forced clean-reagent finding, the template fixed under the complete event view, machine move or whole-template discard or duplication with the right root rules impossible by spine shape, the pool with letter conservation, no firing mints a letter, and the antiparallel law lifting the crown automorphism to all runs of all machines, present and future; CH3 closed as delivered, standalone: the polymerase built as one closed term over the strand's active form, correctness compiled as polRun onto the antisense spine, the jewel polMirror, replication as reflection enacted one-handed, and polTwin closing the duplex, the mirrored twin walking the mirrored presentation back to the sense strand; the two long behavior chains machine-derived by a normalizer and checker-certified, a method line for the engineering log; the chiral line complete through its chartered arc. Movement zero's registration, historical: the recognition analysis, orthogonality cuts both ways, stuck against fired is not a branch, the machine reads its own hand and emits the other as inert cargo, four movements mapped on the C5 playbook with mirrorChain as the correctness shape; movements one through four registered. CH3 sharpened: cross-hand recognition is a new physical channel, hybridization as rule extension, design point open. Checker disciplines three and four logged: shape views for function-valued indices; declare transported implicits. Newly opened (line W, the witness's silhouette, per the Part Three concept note): W1, the no-quoter theorem, no citizen sings an arbitrary colleague's song, proof by the identity-applied-to-itself collision with confluence cited at the joint or determinism against the engine, the centerpiece candidate; W2, the provenance lemma, every occurrence in a result traces through firings to the inputs, over the occurrence ledgers; W3, the registration-costs-bonds family behind the purity fact, any record costs at least one condensation; W4 closed as delivered, engine tier: the stochastic recruit demonstration, W4.idr self-contained with deterministic seeds, eight histories of six distinct shapes arriving at one normal form with the verdict line naming gardenUNF, and the knot wandering

fuel-out as the contrast; the fifth-criterion fence in the note's first paragraph, the rhyme marked as rhyme, the illustration pointing at receipts and never replacing them. Line W is complete, three theorems and one illustration. None blocks the drafting of chapter eleven; W1 before its final audit. Newly opened with the book's Part Three restructuring and the physics chapter: P1, typed confluence via Newman, strong normalization plus local confluence on the typed fragment yields confluence on the garden, compilable with existing machinery, upgrading the arrival joint from cited-classical to compiled-on-the-garden for chapters six, eight, and thirteen; D1 closed as delivered, packaged: the tagged choice compiled in module Choice from the certified B chain and a four-move thrush, leftTag and rightTag with explicit eliminations, the dictionary's or-row armed at the object level with the meta level noted, Either load-bearing since rootStatus; and the standing note that the quantum gravity testimony of chapter thirteen claims nothing compiled beyond stepsNeverMint and the fragments already delivered, per the Senf discipline. Tracking: concordance v0.3 delivered as the book-to-module audit trail; refresh it at every chapter delivery and every module campaign close. M1 closed at the foundation per the working note: a ten-module package, Core through Quote plus the extracted Values module, builds from scratch clean in under two minutes, covering lines 1 to 2,050 and the Val datatype; part S is a leaf outside the chain by finding, and the upper region keeps its monolith verification status. W1 closed as delivered, packaged: the no-quoter theorem compiled as the package's eleventh module in the fast lane, parametric over any normal-valued separating coding, with uniqueness of normal forms carried as the explicit classical hypothesis, Church and Rosser cited at exactly this joint; the formalization's finding, that the program's own quote is behavioral rather than normal-valued, so the C5 reader evades rather than contradicts the theorem, is registered for chapter eleven and the receiver chapter. P1 closed as delivered, packaged, beyond its registration: local confluence of the untyped calculus compiled whole in module LocalConf, head-inversion helpers and a disciplined peak analysis, and Newman's lemma compiled in module Newman with strong normalization as a local accessibility predicate, yielding unfAt, uniqueness of arrival at every terminating point; consequence, confluence is no longer cited anywhere in the program, only termination is, and W1 is upgraded to noQuoterSN, the global hypothesis replaced by one termination certificate for the quoter's own run. W2 closed as delivered, packaged: the provenance family compiled in module Provenance, stepNoMint and stepsNoMint, absence hereditary along every walk with the Starling case showing why copying is not minting, provenance as the contrapositive, and custody localizing the force at a meeting, a marker in an outcome entered through one of the two parties; the receiver chapter's formal floor, with the semantic-aboutness boundary marked in the paper. W3 closed as delivered, packaged: the registration-costs-bonds family compiled in module Purity, the silence ladder from zeroBondsAtomic through registrationCosts, any change presupposes a standing bond, and the two firing bills exact, the Kestrel erasing two bonds and a ledger whole, the Starling's entire growth one copy of the shared ground, the bond-level twin of provenance; the purity fact itself remains a design truth, its countable shadow now priced. P1b closed as delivered, packaged, by re-proof rather than composition: strong normalization of the typed garden compiled whole in module GardenSN, the reducibility method with the blowup block replaced by the Sat column machine, satRed and satCol, four views and two closures where the original carried twenty-one dependent clauses; derivations Church-encoded as DedC after the checker anomaly was minimally reproduced, a three-clause Nat-valued match on the derivation family rejected as non-covering while Step and Sat match cleanly, M1b's reproduction now in hand with the named discipline, Church-encode what the checker will not split; gardenSN, gardenConfluence, and gardenUNF delivered with zero citations, and noQuoterGarden closing W1 end to end for typed quoters, nothing classical anywhere in the chain. Part S's content is no longer hostage to any vehicle. The original P1b registration, superseded: the six-line garden corollaries composing newman with strongNorm await an S-context vehicle, blocked by the environment find-

ing, now triple-confirmed, that part S elaborates only inside the full monolith, control slice green, minimal S slice killed, warm 2,229-line prefix killed, historical full check to 3,644 standing; M1b's reproduction brief is enriched by these three data points. M1b closed as documented and upstream-linked per the report: anomaly A, the coverage knife-edge, minimally reproduced in thirteen lines, CoverageRepro.idr, with a twenty-two-variant bisection matrix locating the joint trigger, a repeated index variable plus a distinct-constant partner with a depth-three arrow index, upstream candidate issue 402 by title; anomaly B, the elaboration space blowup, reproduced in MemoryRepro.idr at 244 lines with the context variance documented, upstream candidate issue 2341 by title; both workarounds proven in production and codified as disciplines, Church-encode what the checker will not split, restyle what the elaborator will not afford. Filing upstream is left to the author with the files ready. Historical registration: M1b, distill minimal reproductions of the two module-context anomalies, the part S elaboration blowup and the EyeOne conversion failures, both documented with artifacts in the M1 note, candidate upstream reports. Environment disciplines registered: swap exists but makes the collector lazy, cap it against the disk; check from the directory matching the module name; long builds are polled within a turn. Newly opened (line C9, exploratory, tier C): the gentle chemistry, BCI or BCIW as primitive atoms, the no-erasure vessel with valence table one, two, three, three and a gentler burn table, kin to the two refusals; pairing non-uniqueness with four letters noted; opened, not pursued. A book chapter follows only if the mathematics delivers. Newly opened per the author (line C7, the between): space as the form of possible meetings, in three tiers graded by difficulty. Tier one, tractable and compiled-targetable: the tree metric on term positions with the metric axioms proved, and the locality theorem, every step rearranges only its redex neighborhood, with S recruitment isolated as the calculus's one nonlocal channel. Tier two, engineering: the geometry dial for the vessel, collisions and recruitment restricted to a neighborhood graph of range r, the phase sweep rerun as r tightens, the well-mixed vessel recovered at infinite r. Tier three, hard and honestly so: derive distance from interaction statistics instead of imposing it, and test whether anything geometric emerges in a well-mixed chemistry at all; the negative outcome, that geometry like structure cannot be minted by reactions and must be put in by the drive, would be reported as a finding, not buried. A chapter, the gap and the meeting place, follows only if tier one compiles and tier two or three yields something worth a reader's evening. Difficulty acknowledged in the registration itself: tier three is the toy version of a problem entire research programs in quantum gravity have held open for decades. The average entropy budget of random terms (the Bendkowski, Grygiel, Zaionc toolkit with this series' weights).

5. Priorities and the book mapping

Shipped with this agenda: N3' s theorem pair (sumStepsNum, sumValue), N4' s boxLen, and L2' s peirceNoProof and dneNoProof, compiled, module at 1,523 lines. Recommended order: L2, C1, C2, L1' s preservation half, and N4 with P1 are delivered; next by leverage: C5' s verified reader (the quotation layer, building on abs), or C3' s verified atlas, or L1' s normalization sequel (one campaign, one paper), then N1 paper-level, with C3 and P3 as parallel short pieces. The long arcs N2, L1' s normalization half, L4, and P2 run behind. Every line lands in a chapter: numbers in The Unreachables, the vessel lines in The Vessel, the coin lines in The Laws of Thought and The Garden, the budgets and the machine in The Price of Forgetting, the boundary in The Witness; and the finale, The Reading, inherits whatever the lines deliver, which is the design.